

1. Write a C program to add / subtract two matrices.
2. Write a C program to check whether two matrices are equal or not.
3. Write a C program to find sum of main diagonal elements of a matrix.
4. Write a C program to find sum of minor diagonal elements of a matrix.
5. Write a C program to multiply two matrices.
6. Write a C program to check Identity matrix. [Identity matrix is a special square matrix whose main diagonal elements is equal to 1 and other elements are 0.]
7. Write a C program to find determinant of a 2x2 matrix.
8. Write a C program to find transpose of a matrix. [Transpose of a matrix **A** is defined as converting all rows into columns and columns into rows.]
9. Write a C program to check Symmetric matrix. [Symmetric matrix is a square matrix which is equal to its transpose.]
10. Write a C program to find upper triangular matrix.
11. Write a C program to find lower triangular matrix.
12. Write a C program to check Sparse matrix. [Sparse matrix is a special matrix with most of its elements are zero. We can also assume that if $(m * n) / 2$ elements are zero then it is a sparse matrix.]